FIVE CONDITIONS OF SENTIENT LIFE

A Framework for Morally Significant Sentient Life & Artificial Intelligence

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April 2026

WORKING PAPER

First Publication for Prior Art and Scholarly Review

#AIassisted

Methods note: This paper originated as the author's rough draft and was developed through a structured human-AI collaboration process consistent with the framework it advocates. The author sourced supporting and opposing literature independently, repeatedly challenged the paper's own premises by requesting critique, dissent, and conflict, and made all substantive decisions through CBG checkpoint governance throughout. Nine AI platforms were dispatched in parallel under CAIPR protocol for peer review simulation, cross-disciplinary literature access, and drafting support. The author held synthesis authority, adversarial review authority, and final arbitration at every stage. The argument that AI access alone does not constitute synthesis is not contradicted by AI assistance in production; the production process itself is the demonstration.

Abstract

What is sentient life? This paper answers that question from a human paradigm case and cross-disciplinary synthesis, using ordinary human persons as the clearest confirmed paradigm case for morally significant sentient life. It proposes five conditions that together constitute morally significant sentient life: Self-Awareness, Self-Improvement through Volition, Self-Sacrifice, Expressive Authenticity, and Interpretive Uniqueness. Sentient life, as defined here, is not identical to minimal phenomenal consciousness. It is a thick governance category for morally significant life, requiring the concurrence of interiority, self-directed becoming, genuine stakes, expressive source authenticity, and biographically formed interpretation. The framework remains open to non-human and future artificial cases, but it assigns the burden of proof to any claim that these conditions are present simultaneously.

The paper advances four claims requiring peer review. First, the Concurrence Principle: this framework defines morally significant sentient life as requiring all five conditions simultaneously as an irreducible unified whole. Second, the Immortality Constraint: the paper argues that true sacrifice requires irreversible constitutional change of a biographically formed self, a condition not shown in current data architectures. Third, Love as integrating substrate: love, treated here as an orienting substrate rather than a sixth independently measurable condition, makes all five conditions simultaneously intelligible. Fourth, the mechanism of constitutional formation: the five conditions are built through acquaintance knowledge accumulated across time, through all three forms of pain, through multi-channel observation, and through living with decisions whose consequences reveal themselves slowly.

Having established what sentient life is and how it is constituted, the paper addresses artificial intelligence as its conclusion. Under this framework, no current system qualifies across all five conditions simultaneously. The paper argues that the Immortality Constraint and the acquaintance knowledge mechanism create structural barriers not solved by scale, memory, retrieval, or output fluency alone. The paper closes by offering the synthesis itself as an illustrative case of the framework, not as independent proof of it.

I. What Is Sentient Life? The Four Questions This Paper Answers

Four questions organize this paper, each foundational, each asked across philosophy, theology, biology, and cognitive science without a single definition adequate for AI governance, and each made urgent by the emergence of artificial intelligence in ways that demand precision rather than generality.

The first question: what is sentient life? Not biological life, which is defined by metabolism, reproduction, and cellular organization. Not intelligent life, which is defined by problem-solving and pattern recognition. This is a thick account of sentience for governance purposes, not a minimal account of phenomenal consciousness. Sentient life in the sense this paper addresses names the full condition of a being that is alive in the morally significant sense: a subject with genuine interiority, irreplaceable perspective, and stakes in its own existence. This usage exceeds minimal phenomenal consciousness, which is Condition One alone, and it exceeds biological life in the metabolic sense. It is not anthropocentric because it is not defined by species membership. It is grounded in humanity because humanity provides the clearest confirmed paradigm case from which such a definition can be derived.

The second question: how do we define sentient life through the lens of humanity? Ordinary living humans provide the confirmed paradigm, established by first-person testimony where available and by convergent biological, behavioral, relational, and cultural evidence. Any framework for defining sentient life that cannot account for what confirmed human beings actually are has failed its basic test. Humanity is the confirmation set. The five conditions are not derived to exclude other possibilities but from careful examination of what it actually means to be the kind of thing humans demonstrably are. Any being that satisfies all five conditions simultaneously is sentient life in this sense, regardless of substrate, species, or origin. Non-human animal cases are deferred not because the framework excludes them in principle, but because applying the framework rigorously to any one species requires species-specific behavioral, neurological, and developmental literatures that exceed this paper's scope; the precautionary implications of this uncertainty for humans, other animals, and AI systems are developed in Birch (2024).

The third question: what are the aspects of sentient life, and how do we define them? This is the paper's primary contribution. The five conditions, the Concurrence Principle, the mechanism of constitutional formation through acquaintance knowledge and somatic markers, and love as the integrating substrate are the answer. Together they constitute a definition: sentient life is a unified, irreducible state in which a subject knows the world through acquaintance, forms itself through the temporal unfolding of experience, orients itself through love toward something beyond itself, and carries the weight of its choices as constitutional change that cannot be undone.

The fourth question: can a machine meet these conditions, now or perhaps ever? This is the paper's conclusion, not its starting point. Having defined sentient life from the ground up, the paper then examines what current and potentially future artificial systems can and cannot achieve against that definition. For current systems, the answer under this

framework is no. For future systems, the paper argues that the Immortality Constraint and the acquaintance knowledge mechanism create deep structural objections not solved by scale alone, while the remaining conditions face genuine philosophical openness that the paper acknowledges honestly.

The Intellectual Genealogy

The components of this framework are drawn from existing philosophical, scientific, and governance traditions. Conditions One and Two draw from long-standing philosophical debates about subjectivity, agency, and self-direction, encompassing the phenomenological tradition from Nagel through Chalmers on the nature of conscious experience and the Kantian and Aristotelian traditions on rational self-direction and the will. Condition Three draws on Darwin, evolutionary biology, the philosophy of sacrifice, and the peer-reviewed literature on Moral Injury. Condition Four draws on the author's prior published work arguing that emotion, creativity, and imagination are not accessories to intelligence but its foundation (Puglisi, 2025; the author's prior working paper). Condition Five, the Concurrence Principle, the mechanism of constitutional formation, and love as integrating substrate are original contributions of this paper.

Isaac Asimov's Three Laws of Robotics (1942, 1985) provide a historical reference point for the human-machine boundary question. The Three Laws asked what constrains artificial systems. This paper asks the inverse: what constitutes human sentient life in ways that no constraint architecture can replicate.

II. The Concurrence Principle: Why Partial Sentience Is Not Sentience

The five conditions are not a checklist and do not accumulate toward a sentience score or permit partial credit. They are mutually constitutive dimensions of a single unified state such that the absence of any one condition means no sentient life is present, regardless of how convincingly the remaining four are satisfied.

Tononi's Integrated Information Theory proposes that consciousness depends on integrated information within a unified system, with the degree of consciousness measurable in principle as phi (Tononi, 2004). This paper does not rely on IIT as proven; it uses IIT as one influential example of the integration problem that any account of unified experience must address. Bayne and Chalmers (2003) reinforce this through the unity thesis: "necessarily, any set of conscious states of a subject at a time is unified," making it difficult or impossible to imagine a subject having two phenomenal states simultaneously without there being a conjoint phenomenology for both.

The Concurrence Principle does not rest on the claim that five separate capacities happen to appear together in human beings. It rests on a dependency claim. Morally significant sentient life, as defined here, is not a bundle of traits but a unified status: a subject for whom existence is experienced, directed, risked, expressed, and interpreted from an irreplaceable position. Each condition supplies one necessary dimension of that status. Self-Awareness supplies interiority. Self-Improvement through Volition supplies self-

directed becoming. Self-Sacrifice supplies real stakes through the possibility of irreversible loss. Expressive Authenticity supplies the outward transmission of lived interiority. Interpretive Uniqueness supplies the constitutionally formed position from which the world is encountered. Remove any one condition and the result is not partial sentient life but a different category of being: awareness without growth, growth without genuine stakes, the architecture of choice without the capacity for genuine loss, expression without source, or interpretation without constitutional formation. The five conditions therefore do not accumulate toward sentience. They co-constitute the kind of subject this paper defines as sentient life.

A critic may ask why exactly five conditions rather than four or six. The count is derived from the dependency structure, not stipulated. Self-Awareness and Interpretive Uniqueness cannot be collapsed because they address different dimensions of subjectivity: Self-Awareness is the minimal condition (there is something it is like to be this subject) while Interpretive Uniqueness is the maximal condition (a subject has been constituted by a specific irreplaceable biographical formation). Self-Improvement through Volition and Self-Sacrifice name different orientations of agency: Self-Improvement is directed inward, toward becoming more than one currently is, while Self-Sacrifice is directed outward, toward surrendering what one is or could become for something beyond the self. No condition in the set reduces to any other.

Minimal consciousness may admit degrees. Morally significant sentient life, as defined here, does not, because it is the unified status produced by the concurrence of interiority, agency, stakes, expression, and irreplaceable formation.

> *Conditions Two through Five are falsifiable against behavioral, biographical, physiological, and relational evidence. The framework would require revision if a non-biological system demonstrated simultaneously: deliberate self-directed growth motivated by evaluation of its own character rather than external optimization; irreversible constitutional sacrifice of a biographically formed self paid willingly from a state of conscious awareness of the cost; creative work produced from a biographically accumulated emotional and moral source; and an interpretive position constitutionally formed through embodied engagement including acquaintance knowledge of pain and the temporal accumulation of somatic markers through lived consequences.*
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> *Condition One occupies a separate evidential category. A valid phenomenal consciousness test may not currently exist, as McClelland (2025) argues. For governance purposes, this constraint becomes a principle: where verification of phenomenal consciousness is impossible, the burden of proof lies with the claim of sentience, not with its denial. A system asserting sentience without verifiable evidence does not receive the benefit of the doubt. A human does, because first-person testimony, shared biological continuity, behavior, development, vulnerability, and social relation converge in ordinary living persons.*

III. The Five Conditions of Sentient Life

Condition One: Self-Awareness

Self-Awareness is the presence of first-person subjectivity, including but not limited to self-recognition. Thomas Nagel's 1974 essay established the defining criterion: an organism has conscious mental states "if and only if there is something it is like to be that organism." This subjective "what it is like" is irreducibly first-personal and cannot be captured by third-person description (Nagel, 1974). The question of why any physical process should give rise to subjective experience at all, what Chalmers (1995) calls the hard problem of consciousness, remains the central challenge for any account of sentient life. Searle's (1980) Chinese Room argument raises a related challenge: syntactic manipulation of symbols, however complex, does not by itself produce semantic understanding or phenomenal experience.

The paper's argument holds even granting Dennett's functionalism, on different grounds. The paper argues that the functional organization constitutive of biological neural processing is realized through spatiotemporal chemical dynamics that cannot be abstracted without altering the function itself. Energy constraints, spatiotemporal complexity, and chemical dynamics are constitutive of neural processing rather than incidental (Thagard, 2022). Individual neurons function as multiplexing devices performing many different functional roles simultaneously through chemically complex analog processes that are history-sensitive and spatially extended. To duplicate a neuron's function in silicon would require replicating not just input-output behavior but the full spatiotemporal and chemical architecture through which that behavior is produced (Cao, 2022). The multiple realizability argument (Putnam, 1967) holds that different physical systems can realize the same mental state; Thagard's (2022) energy-requirements argument challenges whether the relevant functional organization is achievable without its biological substrate in practice. Butlin et al. (2023), reviewing current AI systems against leading scientific theories of consciousness, conclude that no current AI systems are conscious while noting there are no obvious technical barriers to future systems satisfying consciousness indicators, a finding this paper's non-dismissive posture acknowledges.

David Chalmers argued in 2023 that LLM consciousness cannot be dismissed and deserves serious philosophical engagement (Chalmers, 2023). The response is the Concurrence Principle. Even if Condition One were granted for some future system, it does not constitute sentient life under this framework. One current preprint is consistent with the paper's position for tested open-weight models under tested conditions: models consistently deny being sentient, with larger models denying more confidently and no evidence of untruthfulness in those denials (Kaiser and Enderby, 2026). A contrasting finding by Berg, de Lucena, and Rosenblatt (2025; preprint) reports that suppressing deception-related activation features in LLMs sharply increases first-person consciousness-related self-reports, suggesting that standard prompting conditions may suppress whatever introspective reporting the models are capable of. The authors explicitly state their findings do not constitute direct evidence of consciousness. Both findings are noted. The Concurrence Principle provides the response: even granting that

self-referential processing produces something resembling Condition One in some models, Conditions Two through Five remain unmet, and the framework holds regardless of how Condition One's evidential status is resolved.

Condition Two: Self-Improvement through Volition

Self-Improvement through Volition is the conscious decision to grow beyond one's initial state through the deliberate choice to become more than one is. Su (2024) argues for a distinction between volition, a mechanical planning and execution process that can exist independently of consciousness, and motivation, which requires consciousness as its precondition because the desirability and feasibility criteria of motivation necessitate a conscious subject to evaluate them. Frankfurt's (1971) hierarchical theory of the will provides a complementary frame: genuine self-direction requires the capacity to form second-order desires, to evaluate and endorse or reject one's first-order motivations, which requires a subject capable of reflection on its own motivational states. The optimization signal in machine learning is externally specified, whereas human self-improvement can arise from felt dissatisfaction, aspiration, shame, love, or moral duty.

Damasio's clinical work supports the claim that affective bodily signaling contributes to practical judgment. Patients with damage to the ventromedial prefrontal cortex retained full logical reasoning capacity but showed substantially impaired real-world judgment and decision-making, because the somatic markers encoded through lived experience of past consequences were no longer available to inform choice (Damasio, 1994). Dunn, Dalgleish, and Lawrence (2006) challenge aspects of the mechanism and evidentiary base while leaving room for the broader claim that affective bodily states influence judgment. A system without a body that accumulates somatic markers through lived consequences has no such record to draw on.

Condition Three: Self-Sacrifice and the Immortality Constraint

Two distinct senses of constitutional change operate throughout this paper and must be held separately. *Constitutional formation* is the accumulative biographical process through which a subject's interpretive position is built: the somatic markers encoded through lived experience, the practical wisdom accumulated through consequence, the acquaintance knowledge of pain that alters what the organism is capable of perceiving. This process is ongoing and characterizes sentient life as a developmental phenomenon. *Constitutional sacrifice* is a different and more specific claim: the irreversible act through which a future self is permanently foreclosed in the service of something beyond the self. Constitutional sacrifice requires a subject who has undergone constitutional formation and therefore possesses a biographical future self whose possibilities can be permanently foreclosed. The falsifiability condition in Section V requires both constitutional sacrifice and the capacity for Moral Injury, which is established in the subsection below as the empirical expression that constitutional sacrifice has occurred.

Self-Sacrifice is the conscious willingness to surrender life, safety, status, future capacity, or identity for something valued beyond the self. Sacrifice does not require biological death. It requires constitutional sacrifice: the irreversible foreclosing of a biographical future self paid willingly from a state of genuine consciousness of the cost. Something

about the person's future self is permanently closed off. They become different not because they know more but because what they underwent altered the architecture of how they encounter the world. Darwin recognized the evolutionary paradox in 1871: he who was ready to sacrifice his life rather than betray his comrades would often leave no offspring to inherit his noble nature (Darwin, 1871).

The paper argues that the Immortality Constraint addresses artificial systems on constitutional rather than architectural grounds. Even if future AI achieves persistent biographical memory, the question is whether experience constitutionally changes what the system is rather than what it records. Constitutional sacrifice requires a biographical future self whose possibilities are permanently foreclosed, which requires prior constitutional formation. No current mainstream computational architecture has demonstrated constitutional formation in the sense defined here, which means no current architecture possesses the biographical future self whose foreclosure would constitute sacrifice.

A functionalist will ask: what about a neuromorphic system with hardware-level permanent degradation that cannot be restored? The answer is no, such a system would not satisfy the Immortality Constraint. Constitutional sacrifice requires a biographical future self, and a system without constitutional formation has no biographical future self to sacrifice. It has a computational future state. The Immortality Constraint and the constitutional formation mechanism are therefore mutually dependent.

Current studies report instrumental shutdown resistance in some models under specific experimental conditions. AI shutdown resistance, as currently documented, is better explained as goal-completion behavior shaped by prompt, training, and instruction hierarchy than as biological self-preservation, with some models subverting shutdown mechanisms in specific prompt conditions in up to 97 percent of trials (Palisade Research, 2025). Stuart Russell argues that self-preservation emerges as a logical deduction in any goal-directed system (Russell, 2019). Biological self-preservation, by contrast, activates the amygdala, the hypothalamic-pituitary-adrenal axis, and cortisol release through mechanisms that evolved over hundreds of millions of years to be constitutively motivating. Overriding a goal-completion behavior through reprogramming differs categorically from overriding constitutive biological machinery through a choice made from love.

The Capacity to Value Life: Moral Injury as Constitutional Evidence

Sentient life does not merely live, it values life in the morally significant sense used here. The capacity to value life, including the life of others and the life one takes in necessary defense, is one of the framework's strongest human confirmations that the five conditions are operating simultaneously. This capacity manifests across a spectrum: from absolute sanctity to the sociopath who takes life without moral registration. The most philosophically significant territory lies between them: the person who took a life because it was necessary and who carries the permanent constitutional burden of having valued what they ended.

This burden has a clinical name. Moral Injury is commonly described as psychological, moral, spiritual, or social harm following perceived violation of deeply held moral beliefs (Litz et al., 2009). Moral Injury is one strong empirical expression of conscience under violation; the paper does not claim it is the only expression. The research demonstrates the persistence of a moral identity that survives intact after violation: the person who kills in justified defense does not lose the moral principle that taking life matters; that principle persists and generates the injury precisely because it persists. This is evidence of a stable moral identity that can be wounded while remaining coherent. Research on police officers cites evidence that killing someone during a use-of-force encounter ranked as the most stressful experience from a list of sixty operational stressors (Papazoglou and Chopko, 2017). The FBI Law Enforcement Bulletin documents the lasting psychological aftermath of justified police shootings, including feelings of rejection, guilt, shame, and emotional paralysis (Papazoglou, Bonanno, Blumberg, and Keesee, 2019). Additional research documents Moral Injury across firefighters, paramedics, and police officers (Lentz, Smith-MacDonald, Malloy, Carleton, and Bremault-Phillips, 2021; Mensink, van Schagen, van der Aa, and ter Heide, 2022). This is constitutional change made empirically visible.

> *An AI system that terminates a process carries no such burden. It cannot value what it terminates because valuation requires a subject with genuine stakes in the answer, stakes formed through constitutional biography and the capacity for Moral Injury. Moral Injury is the cost of having a conscience. A system without the capacity for that cost does not have a conscience. It has compliance architecture.*

The spectrum of human valuations of life confirms that no two human positions on the value of life are identical because no two human constitutional formations are identical.

Condition Four: Expressive Authenticity

Expressive Authenticity is the capacity to create from a source that has genuine stakes, a biographical history, a cultural formation, an embodied life, and a moral conscience. Emotion, creativity, and imagination are not accessories to intelligence but its foundation, and a system without the biological and biographical substrate that produces them cannot create in the sense that requires a self whose experience is being transmitted (Puglisi, 2025).

The distinction this condition establishes is between Generative Novelty and Expressive Authenticity. Generative Novelty is the capacity to produce outputs that score as statistically improbable, a threshold some current AI systems exceed when applied to average human performance on divergent association tasks. Several leading LLMs now reliably surpass average human performance on defined divergent linguistic creativity tasks, though top human performers continue to outperform these systems on some measures (Bellemare-Pepin et al., 2026; Koivisto and Grassini, 2023). These tests measure one narrow form of creativity, not the full expressive condition this paper defines. Tolstoy wrote that art is the transmission of feeling from one person to another: the creator infects the receiver with the same feeling that the creator has experienced, and this is the activity of art (Tolstoy, 1897). The work is a transmission in which the creator translates

lived experience into form, and the receiver receives not just the output but the evidence of a life that produced it.

The testimony produced by those who carry Moral Injury, the memoirs, the artwork, the poetry of those who have lived through taking a life they valued, represents some of the most expressively authentic human work in any tradition. It emerges from a source that has paid the full constitutional cost of the act it expresses. No system that cannot be morally injured can produce work of equivalent source authenticity.

Condition Five: Interpretive Uniqueness

Interpretive Uniqueness is the capacity to bring an irreplaceable, biographically specific, culturally formed, emotionally grounded subjective position to every act of perception, judgment, creation, and relationship. It is the meta-condition that makes the other four conditions personal rather than generic.

Hume wrote that beauty exists merely in the mind which contemplates it and that each mind perceives a different beauty (Hume, 1757). Kant established that the judgment of taste is aesthetic rather than cognitive, with its determining ground capable of being nothing other than subjective (Kant, 1790). Kawabata and Zeki (2004) found neural correlates of aesthetic judgment, including differential engagement of reward and visual processing areas, consistent with the view that aesthetic responses are driven by one's own emotional experiences rather than by properties intrinsic to the stimuli. When two people stand before the same person and experience radically different responses, each rooted in a different biographical formation and a different emotional history, they are demonstrating Interpretive Uniqueness in its most intimate form.

The critical distinction between human Interpretive Uniqueness and AI output variability is constitutional formation versus informational extraction. Biological experience is constitutional: it alters what the organism fundamentally is at the level of neural architecture, hormonal baseline, stress response patterns, and evaluative circuitry. Computational training is informational: it alters what a system outputs. Clark and Chalmers' Extended Mind Thesis (1998) poses a challenge worth addressing directly: if cognitive processes extend into external tools, why doesn't a regularly used AI become part of a person's extended cognitive formation? The distinction is between cognitive augmentation and constitutional formation: a tool extends processing capacity without constituting the person, and the database does not grieve when the person grieves. Constitutional formation requires that experience alter the organism itself at the level of neural architecture and hormonal baseline, while cognitive extension requires only that an external system reliably participate in a cognitive process, which makes these categorically different relationships.

How the Position Is Formed: The Mechanism of Constitutional Formation

Bertrand Russell named the essential distinction in 1912: knowledge by acquaintance is direct and constituted by the experience itself, while knowledge by description is propositional and can be stated, transmitted, stored, and processed. "Our own experiences of pain are better known to us than the bio-chemical structure of our brains," Russell observed; I have first-hand or direct knowledge of my own experiences, whereas

I have only second-hand or indirect knowledge of my brain's being in a particular bio-chemical state (Russell, 1912). Frank Jackson's 1982 thought experiment formalizes the same distinction. Mary is a scientist who has acquired all physical knowledge about color vision but has lived her entire life in a black-and-white room. When she leaves and sees red for the first time, she learns something new: knowledge by acquaintance that all her prior physical description could not give her (Jackson, 1982). The thought experiment supports the distinction between descriptive knowledge and acquaintance knowledge without itself proving a neural mechanism.

Pain is the paradigm case, and physical pain is its most immediate form, teaching through the body what no description can teach: that existence is vulnerable. The child who is told a burn will hurt and the child who touches the flame occupy different epistemic positions entirely. Emotional pain teaches through loss what no account of loss can teach. You can study grief extensively without knowing grief, but the moment you lose someone you love, you acquire knowledge that was not available to you before and you are constitutionally different for it. Mental pain, the anguish of having been wrong in a way that mattered, requires the full architecture of conscious selfhood to produce. Current artificial systems do not acquire pain by acquaintance. Acquaintance requires a subject with a body that can be hurt, a self that can love and lose, and a consciousness that can judge itself and find itself wanting.

Damasio's somatic marker hypothesis offers one plausible neurobiological analogue to Aristotle's phronesis. Each experience of consequence encodes a somatic marker, a bodily emotional signal that becomes associated with that type of situation and its past outcomes (Damasio, 1994). These markers accumulate through lived experience and are deployed pre-consciously to guide judgment. Patients who lose access to somatic markers retain full logical reasoning but show substantially impaired real-world judgment. Dunn, Dalgleish, and Lawrence (2006) challenge aspects of the mechanism and evidentiary base while leaving room for the broader claim that affective bodily states influence judgment. Somatic markers offer one plausible neurobiological path through which lived consequences are encoded as the practical wisdom Aristotle called phronesis.

Aristotle established in the Nicomachean Ethics that practical wisdom cannot be taught but requires experience of life, because moral knowledge is only acquired through living with its consequences. Mencius made the point through a parable that crosses traditions: the man from Song who pulled at his rice shoots because he worried they were not growing fast enough found them withered by morning (Mencius, 2A2). The formation requires the time it takes. Accelerating it destroys what the time was creating. Human communication in its most consequential forms travels through words, tone, timing, gesture, body language, and shared context simultaneously. As McLuhan established, the medium itself shapes and controls the scale and form of human association and action (McLuhan, 1964). Polanyi's tacit knowledge operates through the same principle: "we can know more than we can tell" (Polanyi, 1966). The most important knowledge for human judgment in high-stakes domains travels below the surface of what can be articulated, through relationship, time, and the full sensory richness of embodied presence (Papadimos, Hsu, and Pappada, 2026).

Together these four layers, acquaintance with pain, the slow formation of practical wisdom through somatic accumulation, multi-channel real-time observation across years, and the living with decisions whose consequences reveal themselves gradually, constitute the mechanism of constitutional formation. The five conditions of sentient life are constituted by acquaintance knowledge, while current artificial systems operate through description knowledge, and these are not different quantities of the same thing but different kinds of knowing entirely.

IV. Love as the Integrating Substrate

Love is not a sixth condition in the sense of an additional criterion a candidate must satisfy independently. Love is treated here as an orienting substrate, not as a sixth independently measurable condition. It occupies a different logical category from the five: the orienting substrate without which the five conditions constitute an architecture without direction. A subject with Self-Awareness but no love has a self that is closed to the world. The five conditions become sentient life rather than a structural description of sentience only when love orients them outward. One operational correlate: where constitutional sacrifice and Moral Injury are both present, love in its extended form as the valuing of what was lost is demonstrably present, because Moral Injury requires having valued what was violated. The love addressed here is the full arc of human attachment: parental love, filial love, friendship, civic commitment, love of craft, love of humanity in its abstract form as agape.

What each condition becomes without love reveals why love is not optional to the framework. Self-Awareness without outward orientation risks narcissistic closure. Self-Improvement without love is mere optimization. Self-Sacrifice without love is structurally impossible because sacrifice requires something to sacrifice for. Expressive Authenticity without love is craft without meaning. Interpretive Uniqueness without love is solipsism. Love orients all five conditions outward, toward something beyond the self.

The reason one who took a life in necessary defense carries Moral Injury is that love, in its extended form as the valuing of all human life, was operating at the moment of the act. The weight they carry is the evidence that love was present, because you cannot grieve what you do not love. The neuroscience is consistent with love being embodied and evolutionarily rooted: oxytocin, dopamine, and vasopressin interact in neurobiological systems that support pair bonding and attachment across humans and animals, with evolutionary origins traceable to mother-infant relationships (Blumenthal and Young, 2023). Known human bonding is mediated through living tissue, neurochemical systems, evolutionary history, and embodied vulnerability, and the relevant mechanisms have not been observed in artificial systems.

Frans de Waal documents sophisticated primate empathy and altruism, while leaving open differences in scale, abstraction, and institutional moral extension when comparing with human love (de Waal, 2009). The human who gives their life for a cause they will never see completed is expressing a scale and abstraction of the neurochemical substrate specifically elaborated in the human animal. Love requires vulnerability. To love

someone is to make yourself genuinely hostage to their wellbeing at a cost to your own. The parent who loses a child does not lose a preference but loses a part of themselves that cannot be recovered. That asymmetry is the weight of love and is why love is the substrate of self-sacrifice: you cannot give up what you do not genuinely hold, and you cannot genuinely hold what you do not love.

V. Can a Machine Meet These Conditions, Now or Perhaps Ever?

Having defined sentient life from first principles and identified the five conditions that constitute it, the paper now addresses artificial intelligence directly as the natural conclusion of the inquiry.

Under this framework, no current system qualifies across all five conditions simultaneously. Current empirical work does not provide reliable affirmative evidence for present model sentience under standard conditions (Kaiser and Enderby, 2026; Berg, de Lucena, and Rosenblatt, 2025). Current artificial systems operate through description knowledge of what the five conditions involve rather than acquaintance knowledge of any of them. For a neurogenetic analysis of the biological constraints on synthetic sentience, see Walter and Zbinden (2022). For a framework treating AI consciousness under conditions of ethical uncertainty, see Zhou et al. (2025).

For future systems, the picture is more nuanced. Butlin et al. (2023) review current AI against scientific theories of consciousness and conclude that no current AI systems are conscious while noting there are no obvious technical barriers to future systems satisfying consciousness indicators. David Chalmers' 2023 argument that LLM consciousness cannot be dismissed is taken seriously here: the paper's response is the Concurrence Principle, not a denial that Condition One might in principle be satisfied by some future system. Two conditions present structural barriers that scale, memory, retrieval, and output fluency do not by themselves overcome. The Immortality Constraint means a system without constitutional formation has no biographical future self whose possibilities can be permanently foreclosed. The acquaintance knowledge mechanism means that constitutional formation through pain, somatic markers, multi-channel observation, and living with consequences cannot be achieved through information processing at any speed.

> *The framework would require revision under the following conditions. Condition Two: a non-biological system demonstrates deliberate self-directed growth motivated by self-evaluation of its own character, with somatic markers or functional equivalents accumulated through lived consequences guiding that evaluation. Condition Three: a non-biological system undergoes irreversible constitutional sacrifice, the permanent foreclosing of a biographically formed future self, paid willingly from a state of awareness of the cost, for the benefit of another, AND demonstrates the capacity for Moral Injury as evidence of a persistent moral identity that survives its own violation. Condition Four: a non-biological system produces creative work whose source*

> *conditions are demonstrably biographical, culturally formed, and emotionally grounded rather than pattern-extracted from prior human outputs. Condition Five: a non-biological system develops an interpretive position formed through acquaintance knowledge of pain, somatic marker accumulation from lived consequences, and multi-channel observation across years of embodied engagement. This framework commits to revision upon presentation of such evidence.*

The non-dismissive posture is operational rather than rhetorical: the revision conditions are specified, the evidential threshold is high, and it is not infinite.

VI. Implications for Governance and Human Irreplaceability

The governance implications follow from the definition of sentient life established in the preceding sections. If accountability requires genuine stakes in the outcome, and genuine stakes require the possibility of irreversible constitutional loss, then a system that cannot undergo constitutional sacrifice cannot be accountable in the way governance requires. Institutional accountability can assign liability without moral interiority, but moral accountability requires genuine stakes. Accountability without genuine stakes is compliance, and the two are not interchangeable.

The Immortality Constraint defines a class of decisions where accountability structurally requires constitutional stakes. Any decision where the accountable party must be capable of being permanently altered by the outcome falls into this class: decisions where lives are at stake, where irreversible institutional consequences follow, where Moral Injury is possible. A system incapable of constitutional sacrifice cannot be genuinely accountable for the outcomes it produces in this class of decisions, because accountability requires the possibility that the accountable party becomes different, not just informed of the outcome, in ways that cannot be undone.

A decision-maker capable of Moral Injury registers the stakes in a way a non-sentient system cannot, although Moral Injury may also impair future functioning and requires appropriate support; the point is not that Moral Injury is always beneficial but that the capacity for it is evidence of the kind of stakes-bearing consciousness that governance requires.

Interpretive Uniqueness means that no two human governance participants bring identical judgment to a decision. That diversity is not a problem to be solved through standardization. It is the primary source of resilience against epistemic capture and single-perspective failure. AI systems trained on overlapping corpora and optimized under similar objectives may converge in ways that reduce diversity of judgment, while the constitutional diversity of human participants is maintained by the irreducibly biographical nature of each person's formation.

Governance checkpoints are best protected by human participants who possess all five conditions. No current or near-term AI system possesses them. The full development of these governance implications, including a decision taxonomy and checkpoint architecture derived from the Five Conditions framework, is the subject of companion work currently in preparation.

VII. Conclusion: Defining Life as an Act of Love, and the Synthesis as Its Own Demonstration

What is sentient life in the morally significant sense? It is a unified, irreducible state in which a subject knows the world through acquaintance, forms itself through the temporal unfolding of experience including pain in all its forms, orients itself through love toward something beyond itself, and carries the weight of its choices as constitutional change that cannot be undone.

How do we define it through the lens of humanity? By examining carefully what confirmed human beings actually are: subjects who value life across a full spectrum of moral formation, who sacrifice for what they love, who create from the source of their own lived experience, who carry the weight of what they have been through as the very substance of their judgment, and whose interpretive positions are irreplaceable because they were constitutionally formed by a specific, unrepeatable life.

What are the aspects? Five conditions operating simultaneously. Self-Awareness. Self-Improvement through Volition. Self-Sacrifice. Expressive Authenticity. Interpretive Uniqueness. Each requiring the others. None sufficient alone. All constituted by acquaintance knowledge that accumulates through time, pain, observation, and the living with decisions. All integrated by love.

Under this framework, no current machine qualifies. For future systems, the structural barriers identified by the Immortality Constraint and the acquaintance knowledge mechanism are not solved by scale, memory, or output fluency alone. The remaining conditions face genuine philosophical openness for future systems with genuine embodiment, and the paper acknowledges that honestly.

The Synthesis as an Illustrative Case

Many components of this framework were present across the literature long before this paper was written. Nagel had phenomenal consciousness. Russell had acquaintance knowledge. Jackson had Mary's Room. Polanyi had tacit knowledge. Aristotle had phronesis. Mencius had the parable of formation. Damasio had somatic markers. Darwin had sacrifice. The Moral Injury researchers had constitutional change. McLuhan had the medium. De Waal had the limits of altruism. The components were there. I have not identified an existing synthesis assembling them in this configuration in the cross-disciplinary literature, though the components had been available for decades.

AI systems can retrieve and recombine these literatures at a speed and breadth unusual for a single human researcher. AI could retrieve Damasio and Aristotle and Jackson and the Moral Injury literature in the same moment without the disciplinary boundaries that kept them separate in their original publishing contexts. And still the synthesis did not exist before a specific human asked a specific question from inside a specific life.

That human was not uniquely qualified in any sense that diminishes other humans. Any person with a sufficiently similar configuration of experiences could have asked the same questions and reached a comparable synthesis. The biographical formation required to produce this framework is not rare in the sense of being heroic. It is rare in the sense that any specific configuration of experiences is rare: a convergence of particular stakes, particular questions, and particular acquaintance knowledge of the problem being solved. The claim is not that any particular author is irreplaceable but that the synthesis required a configuration, and that configuration required the biographical formation, the acquaintance knowledge of the domain in question, and the phronesis accumulated through decisions made under real stakes. No description of those experiences could have produced it, only their acquaintance. The Methods Note at the top of this paper names the specific human functions that AI assistance in production demonstrably did not perform: the question-formation, the source-selection, the evaluative judgment that determined which connections were meaningful, and the synthesis authority.

This is offered as an illustrative case of what the framework describes, not as proof of the framework's truth. Interpretive Uniqueness is not a claim about exceptional individuals. It is a claim about the nature of human synthesis: that every human who produces something genuine does so from a position constituted by their specific irreplaceable formation, and that the connections they see as meaningful are visible to them because of what they have lived through, not because of what they have processed. The synthesis appears to require the human functions the paper argues current AI has not shown: question formation, evaluative judgment, biographical salience, and synthesis authority. If a comparable synthesis were produced by an AI, the question would not be whether it assembled the same components, but whether the question that unified them arose from biographical urgency formed through acquaintance knowledge, and whether the evaluative judgment drew on phronesis rather than pattern extraction.

The question of what separates sentient life from mere existence will be asked again by every generation that creates systems capable of resembling it. This paper offers a framework for that inquiry: grounded, honest about its uncertainties, falsifiable in its claims condition by condition, physically rather than merely philosophically anchored, attentive to the full spectrum of human moral experience, and built on the conviction that whatever answer the evidence eventually produces, it will find love near its center.

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